

MIT ACADEMY OF ENGINEERING

COURSE CODE: CS104

5 APRIL 2024

FY BTECH SEMESTER-II 2022 (BACKLOG) PATTERN EXAMINATION 2023-2024

DEPARTMENT OF F.Y.BTECH
MID SEMESTER EXAMINATION
Essentials of Data Science

SET
B

TIME : 1 HOUR

MAX MARKS : 20

TOTAL NO OF QUESTIONS: 20

TOTAL NO OF PRINTED PAGES: 4

INSTRUCTIONS TO CANDIDATES:

1. Assume suitable data wherever necessary
2. Non programmable scientific calculators are allowed
3. Black figures to the right indicate full marks

- | | | | | |
|---|---|-----|-----|----|
| 1 | Select correct option for the given question:
y = 10
x = y += 2
print(x) | [1] | CO1 | L2 |
| | a) 12 b) 10 c) Error d) None | | | |
| 2 | What will be printed by the following code when it executes?
i = 1
while i < 6:
print(i)
i += 1 | [1] | CO1 | L3 |
| 3 | What will be printed by the following code when it executes?
sum = 0
values = [1,3,4,8]
for number in values:
if (number%2 == 0):
sum = sum + number
print(sum) | [1] | CO1 | L3 |
| | a) 0 b) 16 c) 12 d) Error | | | |
| 4 | What will be printed by the following code when it executes?
def div(y,x):
return y/x
print(div(3,5)) | [1] | CO1 | L3 |
| | a) 0.6 b) 0.06 c) .6 d) 6 | | | |

- 5 Recursion and iteration are the same programming approach. [1] CO1 L2
- a) TRUE b) FALSE c) Error d) None
- 6 Which operator is used to perform floor division? [1] CO1 L1
- a) % b) // c) / d) **
- 7 What will be printed by the following code when it executes? [1] CO1 L3
- ```
i=0
while i<3:
 j=0
 while j<4:
 print('*',end='')
 j = j+1
 print('*')
 i = i+1
```
- 8 \_\_\_\_ variable is the variable that is stored inside the function. [1] CO1 L1

- 9** What will be printed by the following code when it executes? **[1]** **CO1** **L3**
- ```
def f1():
    x=15
    print(x)
x=12
print(f1())
```
- a) Error b) 12 c) 15 d) 1512
- 10** Which command below closes the already open file myText.txt if the following code has already been written? **[1]** **CO1** **L2**
- ```
ref_file = open("myText.txt", "r")
```
- a) close()                      b) ref\_file.close()  
c) close(ref\_file)                      d) close("myText")
- 11** Which of the following methods can be used to remove an element from a tuple in Python? **[1]** **CO2** **L2**
- a) remove                      b) del  
c) pop                      d) cannot be deleted
- 12** What will be printed by the following code when it executes? **[1]** **CO2** **L3**
- ```
set1 = {1, 2, 3}
set2 = {3, 4, 5}
print(set1 & set2)
```
- a) {1, 2, 3} b) {3}
c) {3, 4, 5} d) {}
- 13** How do you access an element in a 2D list in Python? **[1]** **CO2** **L2**
- a) list[index] b) list[row][col]
c) list[row, col] d) list[element]
- 14** What will be printed by the following code when it executes? **[1]** **CO2** **L3**
- ```
my_dict = {'a': 1, 'b': 2, 'c': 3}
value = my_dict.get('d', 0)
print(value)
```
- 15** What type of error will the following code produce? **[1]** **CO2** **L3**
- ```
my_dict = {'a': 1, 'b': 2, 'c': 3}
del my_dict['d']
print(my_dict)
```
- a) syntaxerror b) keyerror
c) indexerror d) No error

